

Analytical Problem Solving

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1) CIA Triad Scenario Identification:

There were two principles violated:

i) Confidentiality - Gaining unauthorised access to a private company's DB is Happened.

ii) Integrity - Hacker altered the contents / Data which were not meant to.

2) Risk Quantification:

DB value = 30,000 \$

Breach Destroys 40% of its value

Breach occurs twice a year

SLE, ALE to be found

$$SLE = 30000 \times 0.40 = \$20,000$$

$$ALE = 20,000 \times 2 = \$40,000$$

3) Brute Force Attack:

8 Character pwd, 62-character set

Attacker tests 1 billion pwds per sec.

Find Size of keyspace, worst case time

$$\text{Keyspace} = 62^8 = 2.18 \times 10^{14}$$

$$\text{Time} = \text{keyspace} / \text{rate} = 218,340.1 \text{ seconds.}$$

$$\text{In hours} = 60.65 \text{ hours.}$$

4) Risk - Prioritization: $\text{Likelihood} \times \text{Impact} = \text{Risk Score}$

A) Ransomware : 20

Prioritization:

B) Phishing : 15

Ransomware

C) Insider theft : 10

Server Exploit

D) DDos : 9

Phishing

E) Server Exploit : 16

Insider Theft

DDos.

5) Passive vs. Active Threat classification:

- a) Sniffing traffic → Passive : Threatens Confidentiality.
- b) ARP Spoofing → Active : Threatens Integrity. (also Confidentiality)
- c) Long Term Reconnaissance → No Alteration → Passive
↳ Threatens Confidentiality.

6. Threat Intelligence Lifecycle:

- iii) The Request itself is logged as a requirement
- iv) OSINT and threat feeds are pulled
- i) Analysts Correlate Collected logs.
- ii) A report is sent to CISO.
- v) The CISO's followup questions shape next cycle

Sniffing - Intercepting & Monitoring the resource.

OSINT → Open - Source Intelligence

CISO - Chief Information Security Officer

Reconnaissance - checking affected area to gather info

ARP - spoofing - MITM, allows attackers to intercept communication.